

General Terminology

|                        |                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------|
| Descriptive statistics | Summarizing, organizing and describing data.                                                              |
| Inferential statistics | using probability theory to extrapolate conclusions from data.                                            |
| Population             | A collection of all individuals or items of interest.                                                     |
| Sample                 | A subset of the population.                                                                               |
| statistic              | A numerical property of a sample.                                                                         |
| Parameter              | A numerical property of the whole population.                                                             |
| Variable               | A characteristic or measurement of interest for the individuals in a population.                          |
| Data                   | The values of the variables.                                                                              |
| Quantitative variable  | A variable that can be measured numerically.                                                              |
| Qualitative variable   | A variable that can be categorized based on some characteristic or attribute.                             |
| Discrete variable      | A quantitative variable that can take on only a finite or countable number of values.                     |
| Continuous variable    | A quantitative variable that can take on any value in an interval of real numbers.                        |
| Nominal variable       | A qualitative variable that has no natural ordering.                                                      |
| Ordinal variable       | A qualitative variable that has a natural ordering.                                                       |
| Frequency              | The number of times a value or category occurs.                                                           |
| Relative frequency     | Normalized frequency, expressed as a percentage.                                                          |
| frequency table        | A table that displays the frequency or relative frequency of each value.                                  |
| Rounding               | Carry the final answer with one more decimal place than the data with the least number of decimal places. |
| Proportion             | A ratio that compares a part to the whole population.                                                     |

Sampling Methods

There are 2 main types of sampling methods: random and non-random.

|               |                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Simple random | Every individual in the population has an equal chance o being selected.                                                                                   |
| Stratified    | The population is divided into subgroups (strata) based on some characteristic, and a random sample is taken from each stratum.                            |
| Cluster       | The population is divided into clusters, and a random sample of clusters is selected. All individuals in the selected clusters are included in the sample. |
| Systematic    | A random starting point is selected, and then every $k^{th}$ individual is selected from the population.                                                   |
| Convenience   | The sample is selected based on convenience or accessibility.                                                                                              |

Levels of Measurement

|          |                                                                 |
|----------|-----------------------------------------------------------------|
| Nominal  | Nornumerically meaningfull(labels).                             |
| Ordinal  | Can order (star rating).                                        |
| Interval | Distance between values is meaningful.                          |
| Ratio    | Has a definite zero, ratios are meaningful (e.g., height, time) |

General Tips

- $\sum_{i=1}^n i = \frac{n(n+1)}{2}$  •  $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$  •  $\sum_{i=1}^n r^i = \frac{r(1-r^n)}{1-r}$  for  $r \neq 1$  •  $\sqrt{x^2} = |x|, x \in \mathbb{R}$  •  $(\sqrt{x})^2 = x, x \geq 0$  •  $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$
- $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$  •  $e^{i\theta} = \cos \theta + i \sin \theta$  •  $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$  •  $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$  •  $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$
- $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

Stat Formulas

|                          |                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------|
| Interquartile range(IQR) | $IQR = Q_3 - Q_1$                                                                      |
| Range                    | $R = x_{max} - x_{min}$                                                                |
| Percentile Position      | $P_k = \frac{k(n+1)}{100}$                                                             |
| Sample Mean              | $\bar{x} \approx \frac{\sum_{i=1}^n x_i}{n}$                                           |
| Population Mean          | $\mu$                                                                                  |
| Median                   | The middle value of a sorted data set.                                                 |
| Mode                     | The most frequently occurring value in a data set.                                     |
| Range                    | $R = x_{max} - x_{min}$                                                                |
| Sample Variance          | $\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$                                |
| Population Variance      | $\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$                                      |
| Standard deviation       | $\sigma = \sqrt{\sigma^2}$                                                             |
| Z-score                  | $z = \frac{x - \bar{x}}{\sigma}$                                                       |
| Skewness                 | $Sk = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left( \frac{x_i - \bar{x}}{\sigma} \right)^3$ |

Probabilities

A probability quantifies the chance that an event occurs. A probability must respect the kolmogorov axioms listed below.

- $\mathbb{P}(A) \geq 0, \forall A \in S$
  - $\mathbb{P}(S) = 1$
  - $\mathbb{P}(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n \mathbb{P}(A_i)$  if  $A_i \cap A_j = \emptyset$  for all  $i \neq j$ .
- |                         |                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------|
| Experiment              | A process that leads to one of several possible outcomes.                                            |
| Outcome                 | The result of a single trial of an experiment.                                                       |
| Event                   | A collection of one or more outcomes.                                                                |
| Sample space            | The set of all possible outcomes of an experiment. Notatied $S$ .                                    |
| Equally likely outcomes | $\mathbb{P}(A) = \frac{ A }{ S }$                                                                    |
| Expected value          | $\mathbb{E}(X) \equiv \sum_{i=1}^n x_i \mathbb{P}(x_i) = \mu$                                        |
| Conditional probability | $\mathbb{P}(A B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$ , $\mathbb{P}(A)$ given $B$ occurred. |
| Additive rule           | $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$                        |
| Multiplicative rule     | $\mathbb{P}(A \cap B) = \mathbb{P}(A B)\mathbb{P}(B)$                                                |
| Complement inverse      | $\mathbb{P}(A) + \mathbb{P}(A') = 1$                                                                 |

Independent events

- knowing one occurred does not change the chance of the other.
- $\mathbb{P}(A|B) = \mathbb{P}(A)$
  - $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$

Mutually exclusive events

- events cannot occur together.
- $A \cap B = \emptyset$
  - $\mathbb{P}(A \cap B) = 0$

Sample space representations

listing, tree diagram, Venn diagram.

Contingency tables

**Theorem 1 (Inclusion-exclusion principle)** For any events  $A_1, A_2, \dots, A_n$ ,

$$\mathbb{P}(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n \mathbb{P}(A_i) - \sum_{1 \leq i < j \leq n} \mathbb{P}(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq n} \mathbb{P}(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} \mathbb{P}(A_1 \cap A_2 \cap \dots \cap A_n)$$